MIL-E-1/1589C(NAVY)
26 Aug 1975
SUPERSEDING
MIL-E-1/1589B(NAVY)
2 June 1971
(See note 26)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

1/ TYPE 8237

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Amplifier, pulsed, 2,900 to 3,100 MHz

ABSOLUTE RATINGS:

Parameter:	Ef	if(surge)	tk	I(focus)	epy	epx	ik	Ik	Pi	Du
Unit:	Vac	a	minutes	A	kv	kv	a	mA	kW	---
Maximum:	7.88	54	---	6.0	105	30	68	285	24	0.0042
Minimum:	7.12	---	12	3.8	---	---	---	---	---	---
				(Note 24)	(Note 2)					(Note 4)

ABSOLUTE RATINGS:

Parameter:	Po	tp	prr	VSWR	T (outlet coolant)	Coolant flow (distilled water) GPM		Coolant flow (60/40 ethylene glycol) GPM	
Unit:	kW	μs	pps	---	°C	Collector	Body	Collector	Body
Maximum:	6.72	11	2,000	2.5:1	85	---	---	---	---
Minimum:	---	---	---	---	---	4	0.625	8	?
		(Note 5)			(Note 6)				

ABSOLUTE RATINGS:

Parameter:	Waveguide pressurization (dry air)	Hydrostatic pressure	TA °C	TA °C
Unit:	psia	psig	Operating	Nonoperating
Maximum:	55	200	+85	+85
Minimum:	24	---	-20	-62
				(Note 8)

PHYSICAL CHARACTERISTICS:

Base:	Special (see figure 1)	Input:	Coaxial connector UG-972/U
Mounting:	See note 6	Output:	"S" Band waveguide
Weight:	130 pounds (max)	Dimensions:	See figure 1

TEST CONDITIONS:

Parameter:	Ef	tk	I (focus)	epy	pd	tp	Du(rf)	VSWR	Coolant flow (distilled water) GPM	
Unit:	Vac	minutes	A	kv	---	μs	---	---		
Tolerance:	---	minimum	---	±5	---	±1	---	---	---	
									Collector	Body
Test 1:	7.5	12	Note 24	90	Note 16	10	0.0042	1.2:1	6.0	1.5
Test 2:	7.8	12	Note 24	96	Note 16	10	0.0039	1.5:1	6.0	1.5
				(Note 2)	(Note 3)		(Note 4)			

GENERAL:

Qualification - Not required

1/ Replaces types: VA 811C, L-3742

8237

FSC 5960

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	-					
4003	Pressurizing (waveguide)	9	-	Method A; 55 psia; t = 900	---	---	---	---
---	Hydrostatic pressure	9	-	140 psig; collector and body inlet closed	---	---	---	---
1143	Forced cooling (pressure drop)	-	-	4 GPM to collector 1 GPM to body (distilled water)	ΔP ΔP	6 4	20 20	psia psia
4289	Heater current, nonoperating	-	-	Ef = 7.5 V; tk = 720 (minimum)	If	27	38	A
1256	Electrode current (cathode)	-	1	pd = 0	ik	49.6	58.4	a
4250	Power output	7, 15	1	F = 2.910 to 3,100 MHz	po	---	---	kw
4251	Output power variation (1)	22	1	pd = ±1 dB from the identification plate value: F = 2,980 to 3,100 MHz	Δpo(1)	---	---	kw
4251	Output power variation (2)	10, 22	1	I (focus) = ±2 percent from identification plate value	Δpo (2)	---	---	k
4214	Cathode emission	18	1	pd = 0	Δik/ik	---	10	%
4261	Stability	-	1	pd = 0	po	---	2	w
4006	X-radiation	15, 17	1		---	---	3	mr/hr
	Quality conformance inspection, part 2							
1105	Permanence of marking	-	-		---	---	---	---
4251	Output power variation (3)	23	1	epy = 90 kv +3 to -7 percent	Δ po (3)	---	---	kw
4266	Direct-interelectrode capacitance	14	-		Cak	---	30	pF
---	Internal arc	20	2	F = 2,980 to 3,100 MHz	---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	5, 7	1	Group S (modified)	t	1,000	---	hrs
---	Life-test end point:							
4250	Power output	-	1	2,980 to 3,100 MHz	po	800	---	kw
1031	Low-frequency vibration	11, 19, 25	-	No voltages applied	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 - Continued							
1031	Variable-frequency vibration	11, 19, 25	-	No voltages applied	---	---	---	---
1031	Vibration fatigue	19, 25	-	Ef = 7.5 V	---	---	---	---
1042	Shock, specified pulse	13, 19 25	-	10G; t = 11 ms; no voltages applied	---	---	---	---
1011	Humidity	19, 25	-	t = 96 hours	---	---	---	---
4268	Ion pumps	12, 19	-		---	---	---	---
---	Heater resistance, cold	19	-		---	0.023	0.030	Ohms
---	Waveguide arc	19, 21	1	VSWR = 1.5:1	---	---	---	---

NOTES:

1. Inspection shall be 100 percent.

2. Any short transient spike on the voltage pulse shall not exceed 115 kv in magnitude.

3. The pulse duration (tp) is defined as the time interval between the first and last instances at which the instantaneous amplitude attains 90 percent of the peak amplitude excluding spikes.

4. One scan cycle shall consist of rf pulses of 10, 4.6, or 2.5 μs programmed within the following limits:

   (a) The maximum instantaneous duty cycle (defined as $\frac{d}{D}$, where d = pulse width in seconds and D = time separation in seconds between the leading edges of two successive pulses) shall be 0.011.

   (b) The scan cycle program shall be such that the average duty cycle shall not exceed 0.0042 over the scan cycle which shall have a maximum duration of 30,000 μs.

5. Life tests.

   (a) Short-time life test. The tube to be tested shall be selected at random from the current production lot, in accordance with the double sampling plan specified, and placed on life test for a period of 72 hours. At the end of this period, each tube shall be examined for those parameters listed under life-test end points, using the initial test limits. The tube shall be judged to be passing if it is within those limits, and failing otherwise. The short-time life test is nondestructive, except for any tube which fails to pass. Tubes which pass shall be returned to the lot from which they were drawn. The sample size, acceptance and rejection numbers, are given in the following table.

Sample	Number	Cumulative Number	Not Passing Accept	Not Passing Reject
1st	8	8	0	2
2nd	8	16	1	2

In the event that two lots are rejected in a series of five lots or less, shipments shall be halted and the cause of the poor quality shall be investigated and corrected if possible. Shipments may be resumed on satisfactory evidence that the cause has been found and corrected.

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NOTES:

5. -Continued

(b) Long-time life test. From the first lot satisfactorily passing the short-time life test, and from subsequent lots as necessary, a tube not subjected to the short-time life test shall be selected at random from the lot and placed on life test until failure or until a period two and one-half times the life specified herein. Each tube is operated at the life-test conditions specified herein until it fails to meet one or more of the life-test end points (end of life values). The tube is considered to be a failing tube at this point, the time-to-failure measured. The tube is then replaced with a tube drawn from the current production lot. A tube shall also be replaced if it reaches two and one-half times the life specified herein and shall be counted as a failing tube at this time. The average of the times-to-failure observed, after three or more tubes have completed the life test, shall equal or exceed the life specified herein. In the event that the average observed life fails to equal the life specified herein, shipment shall be halted and the cause of the unsatisfactory performance in the life test investigated. Shipments may be resumed on the satisfactory evidence that the cause has been found and corrected.

6. The TUT shall be mounted with its beam axis within 30 degrees of the vertical. A flexible length of waveguide or other equivalent connection shall be provided between the TUT output flange and load or output waveguide system to prevent excessive forces from being applied to the output flange. The cathode stem shall be immersed in oil as shown on figure 1.

7. The TUT shall amplify frequencies over the band from 2,910 to 3,100 MHz without any change of the electrical or mechanical parameters. Further, in setting up for the quality conformance inspection, part 1 tests, the focus coil currents, the drive power, and other parameters (within their specified ranges), shall be adjusted to the values specified herein. Once adjustment values are set, no further change in values shall be allowed.

8. During storage and extended nonoperating periods, the coolant shall be removed to avoid damage to the tube and cooling paths shall be dried and sealed.

9. There shall be no leakage at maximum limit specified herein.

10. The TUT shall provide the power output specified herein when each coil current is varied ±2 percent of the value specified on the identification plate. All coil current shall be varied simultaneously in the same polarity.

11. The TUT shall be vibrated in each of two planes specified below:

(a) Where the orientation of the TUT has the main axis and major cross section of the TUT elements normal to the direction of the accelerating force.

(b) Where the orientation of the TUT has the main axis of the TUT normal and the major cross section parallel to the accelerating force.

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NOTES: -Continued

12. The Vacion pump, or equivalent, shall have the following ratings:

ABSOLUTE-RATINGS:

Unit:	kV	mA	V(pk ripple)
Maximum:	3.2	10	108
Minimum:	---	---	---

Absolute ratings for the pump apply during all conditions. The pump current is proportional to the voltage applied and the pressure in the pump. The pump is operative when the applied voltage is between 2.0 and 3.2 kV.

13. This test shall be conducted prior to all electrical tests. The TUT shall be subjected to a total of 20 shocks; five shocks in each of the planes specified below:

(a) Where the orientation of the TUT has the main axis and major cross section of the TUT elements normal to the direction of the accelerating force.

(b) Where the orientation of the TUT has the main axis of the TUT normal and major cross section parallel to the accelerating force.

(c) Where the orientation of the TUT has the main axis parallel to the accelerating force and the base down or toward the hammer.

(d) Where the orientation of the TUT has the main axis parallel to the accelerating force and the base up or away from the hammer.

14. The anode to cathode capacitance shall be measured with the TUT in the air, without filaments and beam voltage leads connected.

15. Rf power output versus frequency:

Frequency (MHz)	Minimum output power (kw)	Maximum output power (kw)
3,100 to 2,980	1,000	1,600
2,970	446	1,600
2,960	178	1,600
2,950	71	1,600
2,940	35.5	1,600
2,930	25	1,600
2,920	20	1,600
2,910	16	1,600

16. A 316 watt pulse (max) shall be permissible over passband of 2,910 to 3,100 MHz. Once chosen, the drive power over the passband shall be constant and shall be recorded on the identification plate. A specified tolerance of ±0.35 dB will be allowed for instrumentation errors.

17. X-ray shielding shall be sufficient to reduce X-radiation to 3 mr/hr at a distance of 3 feet from any integral part of the tube above the bottom pole piece, except for conical regions whose axes are colinear with the axis of each coolant fitting.

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NOTES: -Continued

18. When operating under cathode current test conditions, Ef shall be decreased from 7.5 to 6.75 V and the cathode emission allowed to stabilize. The percent change in ik shall not exceed the value specified herein.

$$\frac{ik(Ef = 7.5\ V) - ik(Ef = 6.75\ V)}{ik(Ef = 7.5\ V)} \times 100$$

19. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

20. Internal arc shall be the first test following the holding period. The TUT shall be operated continuously for 6 hours. The epy shall be applied at 65 kv (min). The time allowed for the voltage to reach 96 kv (min) shall be 10 seconds (max). Immediately following application of epy and for the duration of the 6-hour operational tests, the peak cathode current shall be monitored at the cathode terminal for peak pulse amplitude variation. During the first 15 minutes of the test, the peak cathode current shall not exceed the normal cathode current by 20 percent more than twice. During the remaining 5-3/4 hours, the peak cathode current shall not exceed the normal cathode current by 20 percent more than once.

21. Each TUT shall be operated continuously for 1-hour minimum. The frequency for this test shall be at the maximum power point as determined in power output. The phase of the VSWR shall be uniformly varied over a 360 degree range at a maximum rate of 6-cycles per hour. Under this condition, the pressure in the output waveguide of the tube shall be reduced to 24 psia (max). The tube shall operate for one-half hour with no evidence of waveguide arcing.

22. Same as note 15 except po = 900 kw minimum for range of 2,980 through 3,100 MHz.

23. Same as note 15 except po = 700 kw minimum for range of 2,980 through 3,100 MHz.

24. The tube focus coil to be used with the TUT shall be as shown on figure 2. The focus coil operating current for each of the six coils shall be between 4.0 and 5.5 A, and shall be recorded on the TUT identification plate.

25. After the completion of these tests, the TUT shall meet the requirements of method 4250 (power output).

26. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

User activities:
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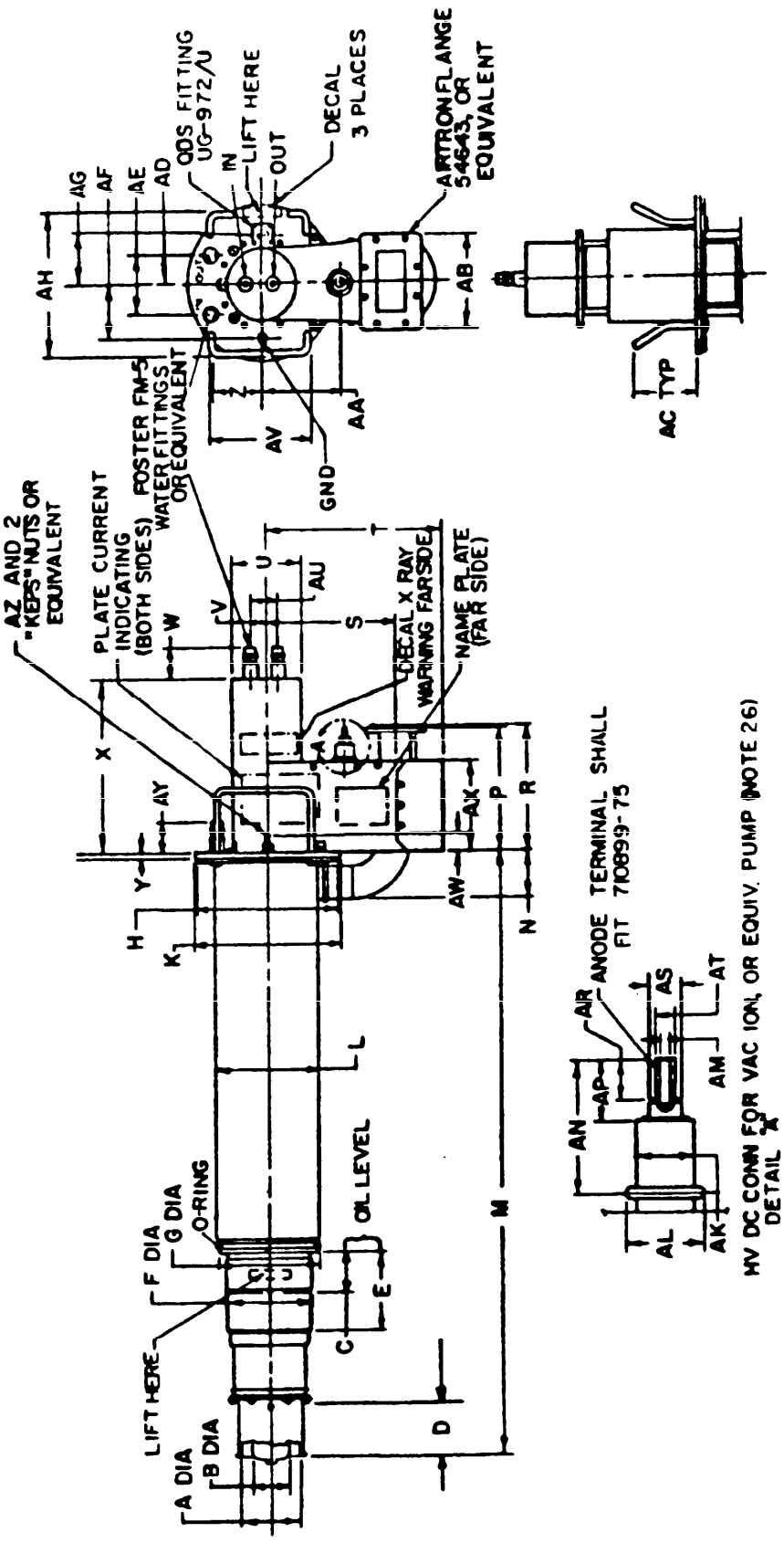


FIGURE 1. Outline drawing of electron tube type 8237.

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Ltr	Dimensions are in inches	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	3.285	3.295
B	2.012	2.022
C	---	2.00
D	1.763	1.783
E	4.00	---
F	4.595	4.655
G	5.347	5.357
M	30.567	30.675
N	---	2.375
P	6.307	6.507
R	6.313	6.563
S	6.922	7.172
T	9.296	9.546
U	3.662	3.712
V	.943	.993
W	1.50	1.73
X	8.925	9.175
Y	.235	.265
Z	2.531	2.655
AA	4.115	4.365
AB	4.735	4.765
AC	3.125	3.625
AD	1.438	1.562
AE	2.938	3.062
AF	2.688	2.812
AG	2.438	2.562
AH	6.687	6.937
AM	.142	.172
AT	.235	.265
AU	1.876	1.996
AV	5.375	5.625
AW	.565	.685
AX	4.469	4.593
AY	1.600	1.680
AZ	.250-28 SST STUD	
Reference dimensions		
H	7.750 Dia.	
K	8.000 Dia.	
L	5.450	
AK	.75 Dia.	
AL	1.03 Dia.	
AN	1.67	
AP	.78	
AR	.53	
AS	.42	

FIGURE 1. Outline drawing of electron tube type 8237 - Continued.

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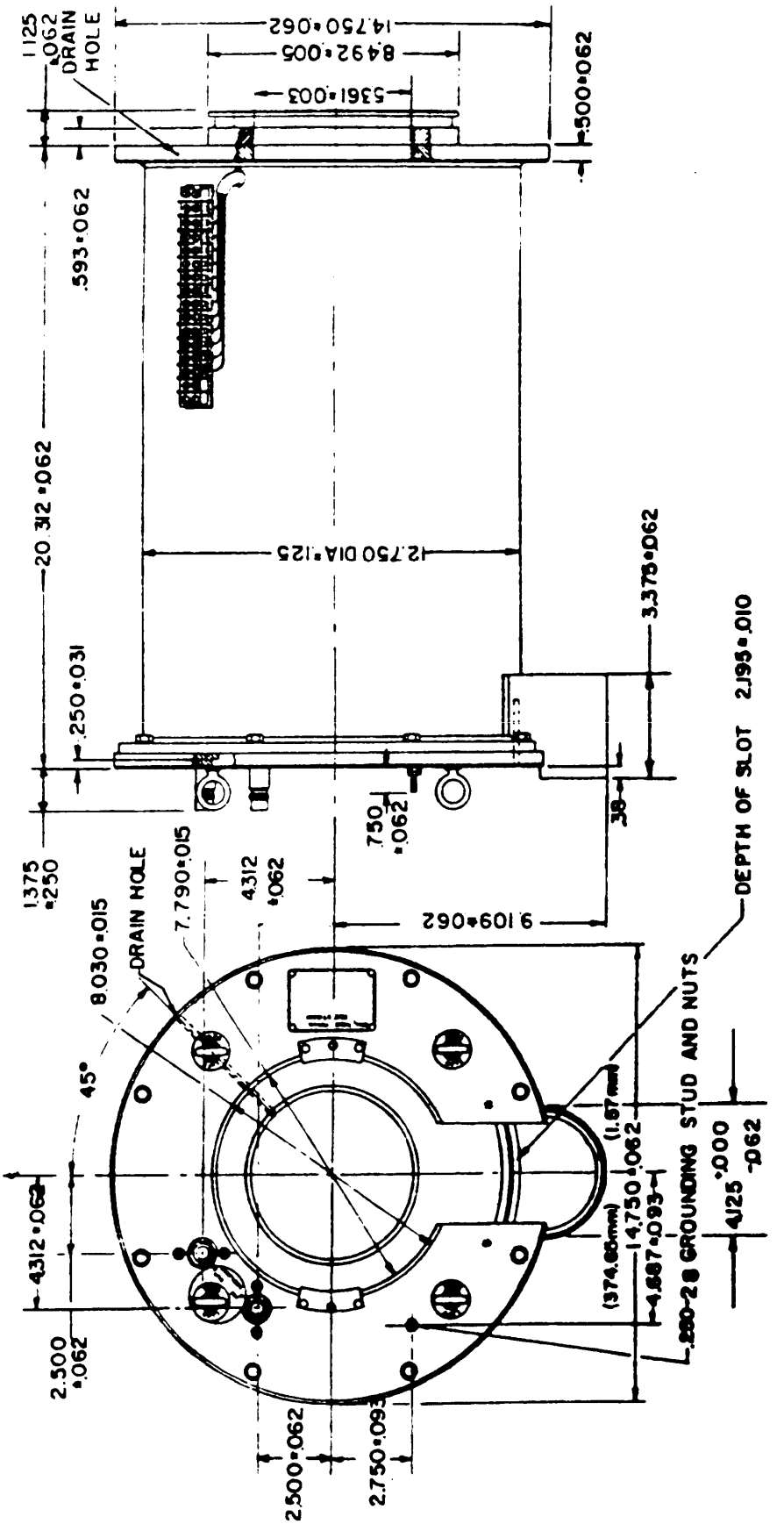


FIGURE 2. Focusing electromagnet.

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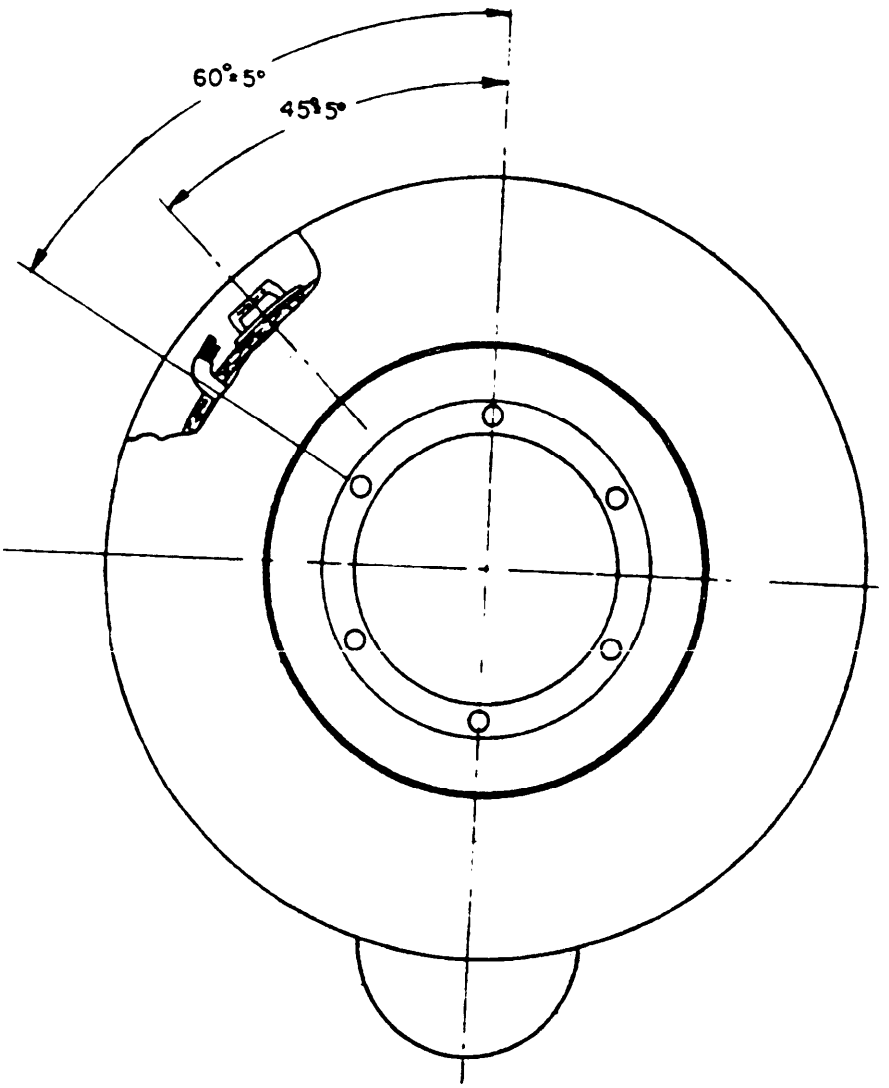


FIGURE 2. Focusing electromagnet - Continued.

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